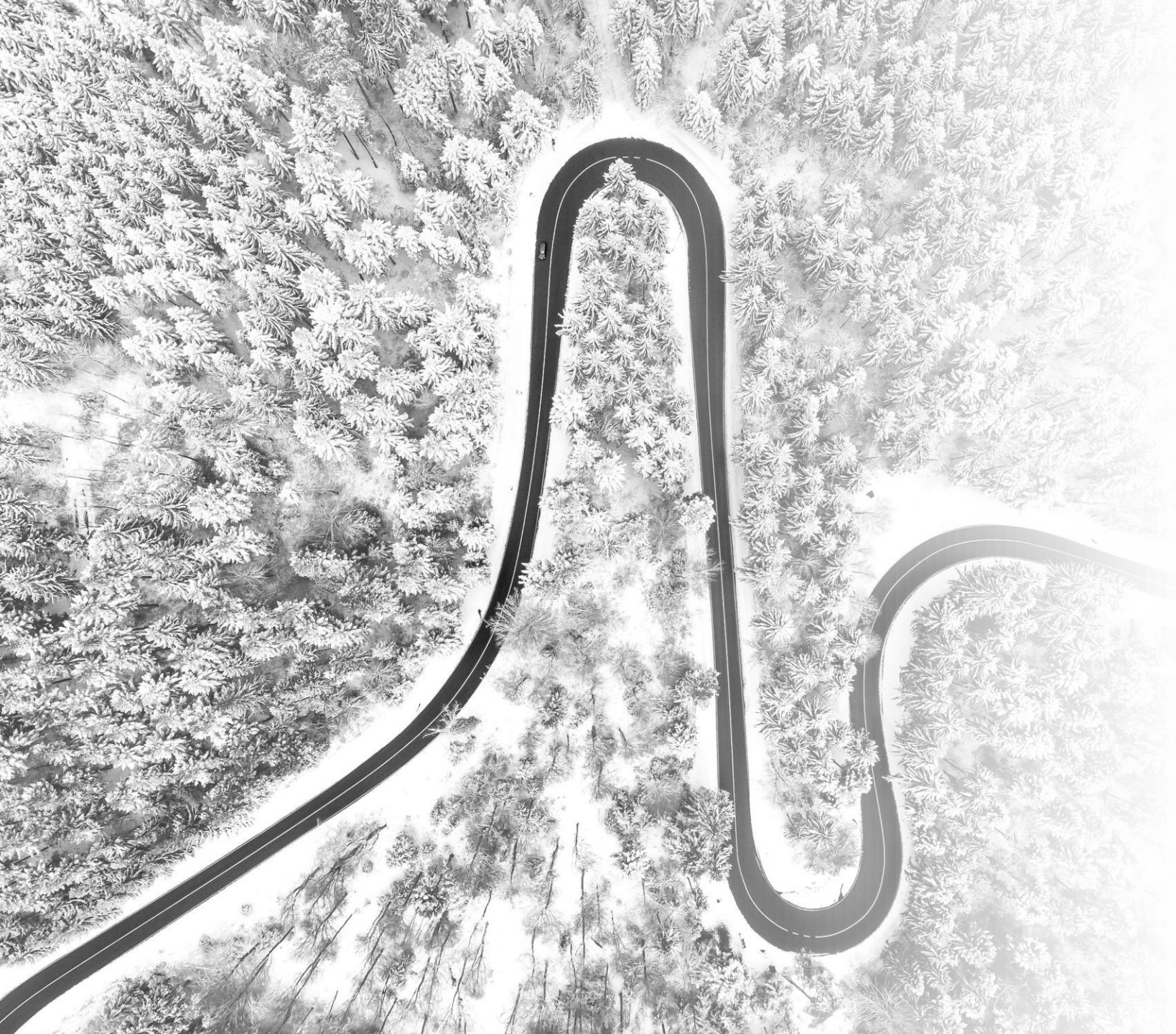


An aerial photograph of a winding road through a dense forest. The road is dark and curves through the lighter-colored trees. The image is partially obscured by a white gradient on the right side.

Deep Learning for Road Network Extraction from HDR Aerial Imagery

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Course: BA-64061-001

Motivation

- **Why Road Extraction Matters**
- Enables autonomous navigation & HD maps
- Provides critical infrastructure analysis for governments
- Helps emergency response during disasters
- Supports logistics, defense, and environmental monitoring
- **Traditional methods fail because:**
- Manual digitization is slow and expensive
- Classical CV (edges, morphology) breaks under shadows, occlusions
- Urban + rural variation is too complex for rule-based systems





Research Objective

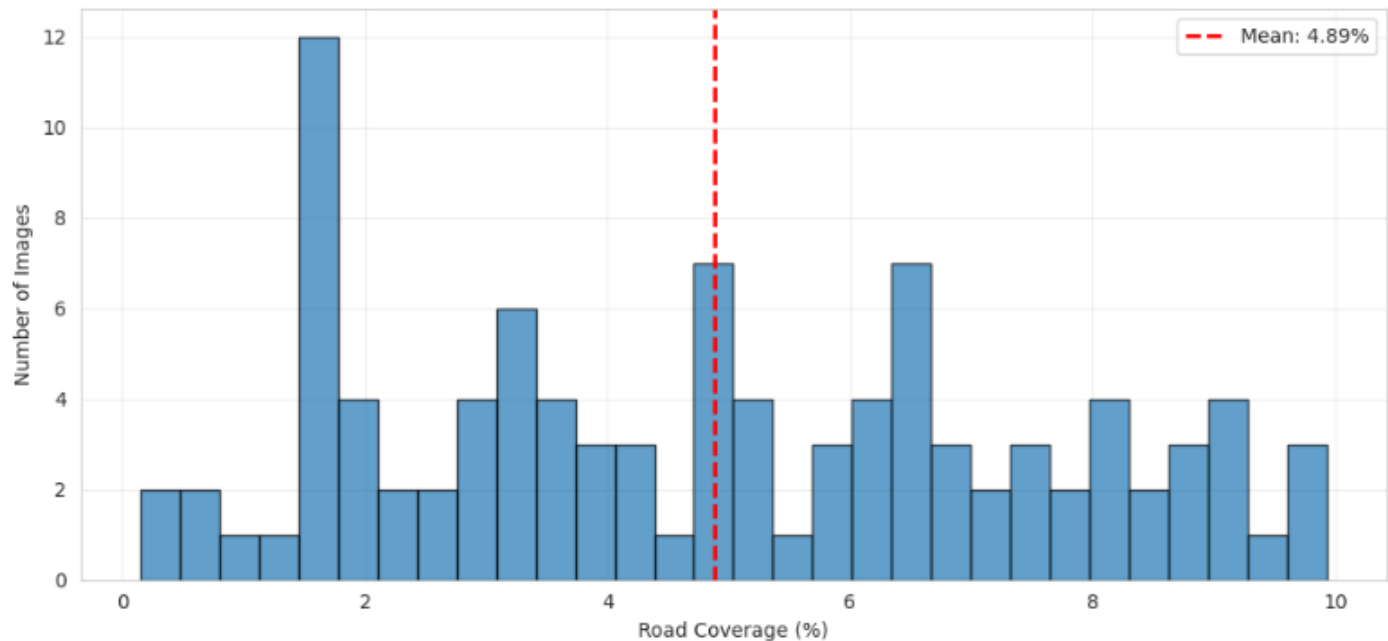
- Evaluate **four state-of-the-art deep learning models** for road segmentation:
- **U-Net**
- **DeepLabV3+**
- **PSPNet**
- **FPN**
- Using the **Massachusetts Roads Dataset**, we compare:
- Pixel-level accuracy
- Road connectivity
- Inference speed
- Architectural strengths & weaknesses

Dataset Description

- **Massachusetts Roads Dataset**
- **1,108** training images
- **14** validation images
- **49** test images
- **TIFFs – 1500×1500 resolution**
- Paired binary road masks
- **Why is the dataset 10GB?**
- High-resolution RGB TIFFs
- Paired masks
- Light compression
- HDR data preserves quality



```
Downloading Massachusetts Roads Dataset from Kaggle...  
Dataset URL: https://www.kaggle.com/datasets/balraj98/massachusetts-roads-dataset  
License(s): other  
Downloading massachusetts-roads-dataset.zip to /content/data  
100% 5.75G/5.76G [00:15<00:00, 956MB/s]  
100% 5.76G/5.76G [00:15<00:00, 403MB/s]  
Dataset downloaded and extracted!
```



Class Imbalance Challenge

- Road pixels = **only 4.89%** of image.
- This causes:
 - Roads disappearing during downsampling
 - High false negatives
 - Bias toward background
- **Solutions used in this study:**
 - Weighted BCE
 - Dice loss
 - Combined BCE + Dice



Methodology Overview

- **Preprocessing Steps**

- Resize to 256×256
- Normalize with ImageNet statistics
- Convert to tensor
- Binarize masks

- **Data Augmentation**

- Flips, rotations
- Brightness / contrast
- Hue / saturation
- Gaussian noise / blur

Model Architectures Compared



- **U-Net**
 - Encoder-decoder
 - Skip connections
 - Fastest + most accurate in this dataset
- **DeepLabV3+**
 - Dilated convolutions
 - ASPP → multi-scale context
- **PSPNet**
 - Pyramid pooling → global scene understanding
- **FPN**
 - Multi-resolution feature pyramid

Training Configuration

- **Optimizer:** Adam
- **LR:** $1e-4$
- **Epochs:** 30
- **Batch Size:** 8
- **Loss:** Weighted BCE + Dice
- **Early stopping** on validation IoU
- **Same pipeline for all models** to ensure fairness



Quantitative Results

- **Key Findings:**
- **U-Net is best overall**
- IoU = **0.2954**
- F1 = **0.4556**
- FPS = **1548.96**
- DeepLabV3+ and PSPNet:
 - Better multi-scale features but struggle at 256×256
- FPN: good recall, lower precision

TEST SET RESULTS (ALL MODELS)

Model	IoU	F1 Score	Precision	Recall	Inference Time (s)	FPS	Parameters
U-Net	0.2882	0.4465	0.2989	0.8897	0.0097	826.60	31,037,633
DeepLabV3+	0.1821	0.3078	0.1941	0.7481	0.0137	585.71	22,437,457
PSPNet	0.1347	0.2374	0.1386	0.8276	0.0066	1205.93	21,437,985
FPN	0.1877	0.3158	0.1986	0.7724	0.0126	634.67	23,155,393

Comparison table saved to: /content/checkpoints/model_comparison_all_models.csv

BEST MODELS SUMMARY:

Best IoU : U-Net

Best F1 : U-Net



Model Strengths & Weaknesses

- **U-Net**
 - Best IoU, F1
 - Fastest inference
 - Loses very thin roads under extreme downsampling
- **DeepLabV3+**
 - Strong multi-scale context
 - Weak performance on thin roads
- **PSPNet**
 - Excellent global context
 - Blurs edges
- **FPN**
 - Good recall
 - Fragmented predictions

Discussion

- **Why U-Net performed the best:**
 - -Skip connections help preserve fine structures
 - -Fewer downsampling layers
 - -Works well even at reduced resolution
- **Dataset constraints:**
 - -1500 → 256 resizing destroys thin road geometry
 - -Severe class imbalance
 - -Shadows and vegetation occlusion
- **Baseline vs published research:**
 - Benchmarks in papers (60-70% IoU) require full-resolution training, not 256×256



Future Work

- Future improvements:
- **Transformer models** (SegFormer, Mask2Former)
- **Topology-aware loss functions**
- **Graph Neural Networks**
- **Multi-resolution training**
- **Domain adaptation** to other regions
- **LiDAR + SAR** multimodal fusion
- Insert an illustration if desired.



Conclusion

- U-Net provided the best balance of **accuracy**, **speed**, and **continuity**
- Thin roads + topology remain major challenges
- Future research will benefit from transformers, multi-modal data, and connectivity-preserving methods
- This study establishes a practical baseline using standardized training across four modern architectures





Thank You
